# Concept Map: Definitions and Results

## Contents


Configuring R . . . . . 1

**1 Dependency diagram** **2**

**2 Descendants of each result** **2**


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### Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
```

```
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

This appendix shows how the definitions and results[1] (`def`, `thm`, `lem`, `cor`, `prp` callouts) in the notes depend on one another.

We say result $B$ is a **descendant** of result $A$ if $A$ is referenced inside the statement or proof[2] of $B$ (directly, or transitively through a chain of intermediate results). The more descendants a result has, the more of the rest of the notes rests on it — so descendant count is a rough measure of how foundational a result is.

The dependency graph is built by `data-raw/callout-graph.R`, which scans the source `.qmd` files and saves the result to `inst/extdata/callout-graph.rds`. This appendix reads that saved file rather than re-scanning the notes on every render, so **re-run that script whenever divs are added, removed, or re-titled** to refresh the diagram and Table 1.

```
library(igraph)
library(ggraph)
library(ggrepel)

cg <- readRDS(here::here("inst/extdata/callout-graph.rds"))

type_levels <- c("def", "thm", "lem", "cor", "prp")
type_labels <- c(
  def = "Definition", thm = "Theorem", lem = "Lemma",
  cor = "Corollary", prp = "Proposition"
)
type_palette <- c(
  def = "#1b9e77", thm = "#d95f02", lem = "#7570b3",
  cor = "#e7298a", prp = "#66a61e"
)
```

There are 271 labeled definitions and results in the notes, connected by 129 direct dependency links.

# 1 Dependency diagram

The dependency diagram is only available in the HTML version of the notes[3].

# 2 Descendants of each result

Table 1 lists every result that has at least one descendant, sorted by the number of total descendants (direct or transitive).

[1] https://morrison-lab.github.io/mds/proof-writing.html#def-theorem
[2] https://morrison-lab.github.io/mds/proof-writing.html#def-proof
[3] https://morrison-lab.github.io/rme/chapters/concept-map.html

```r
ranked <- cg$nodes[cg$nodes$n_desc >= 1, ]
ranked <- ranked[order(-ranked$n_desc, ranked$id), ]

descendant_table <- data.frame(
  Result               = ranked$title,
  Type                 = type_labels[ranked$type],
  `Direct descendants`  = ranked$n_direct,
  `Total descendants`   = ranked$n_desc,
  check.names = FALSE, row.names = NULL, stringsAsFactors = FALSE
)

knitr::kable(descendant_table, format = "pipe", align = "llrr")
```

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result | Type | Direct descendants | Total descendants |
|---|---|---|---|
| Odds | Definition | 4 | 14 |
| Odds as a function of probability | Theorem | 4 | 13 |
| log-odds | Definition | 3 | 10 |
| Potential mean | Definition | 2 | 8 |
| Derivative of odds function | Theorem | 3 | 8 |
| Probability as a function of odds | Theorem | 4 | 8 |
| Causal marginal risk | Definition | 5 | 7 |
| Derivative of log-odds by odds | Lemma | 1 | 7 |
| Conditional predicted risk | Definition | 3 | 6 |
| logit function | Definition | 3 | 6 |
| Derivative of odds w.r.t. log-odds | Lemma | 2 | 6 |
| Derivative of log-odds by odds | Theorem | 1 | 6 |
| First-order interval failure probability | Definition | 3 | 5 |
| One-way random-effects model | Definition | 5 | 5 |
| Risk-set score total | Definition | 2 | 5 |
| Causal marginal risk under identification assumptions | Theorem | 3 | 5 |
| Derivative of log-odds by probability | Theorem | 2 | 5 |
| G-computation | Definition | 3 | 4 |
| Hazard function, hazard rate, hazard rate function | Definition | 3 | 4 |
| Derivative of odds in terms of probability | Theorem | 1 | 4 |
| Derivative of inverse odds function | Theorem | 1 | 4 |
| Proportional-hazards decomposition of the hazard | Theorem | 1 | 4 |
| expit, logistic, inverse-logit | Definition | 2 | 3 |
| inverse odds function | Definition | 2 | 3 |

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result | Type | Direct descendants | Total descendants |
|---|---|---|---|
| Proportional-hazards first-order interval failure probability | Definition | 1 | 3 |
| Cox proportional hazards model | Definition | 1 | 3 |
| Point-mass optimality of the profile baseline hazard | Lemma | 1 | 3 |
| Derivative of probability w.r.t. log-odds | Theorem | 3 | 3 |
| Marginal risk as a conditional expectation | Theorem | 3 | 3 |
| Odds via the odds function | Corollary | 1 | 2 |
| Cox-Snell generalized residuals | Definition | 2 | 2 |
| Leverage for a GLM | Definition | 2 | 2 |
| Observed marginal risk | Definition | 2 | 2 |
| Odds function | Definition | 1 | 2 |
| Odds ratio | Definition | 2 | 2 |
| Pearson residual | Definition | 1 | 2 |
| Cox PH partial likelihood | Definition | 2 | 2 |
| Risk Score | Definition | 1 | 2 |
| Derivative of log-odds with respect to coefficients | Lemma | 2 | 2 |
| Simplified expression for inverse odds function | Lemma | 1 | 2 |
| Odds from log-odds | Lemma | 1 | 2 |
| One minus inverse-odds | Lemma | 2 | 2 |
| Profile likelihood in terms of point masses | Lemma | 2 | 2 |
| Single-subject failure probability in a risk set | Lemma | 2 | 2 |
| Consistency of g-computation | Theorem | 2 | 2 |
| Inverse-odds function inverts the odds function | Corollary | 1 | 1 |
| Akaike Information Criterion (AIC) | Definition | 1 | 1 |
| Collapsibility | Definition | 1 | 1 |
| conditional hazard | Definition | 1 | 1 |
| Deviation of an observation from its subpopulation mean | Definition | 1 | 1 |
| Difference in covariate patterns | Definition | 1 | 1 |
| Difference in log-odds | Definition | 1 | 1 |
| Estimated Risk Score | Definition | 1 | 1 |
| Hat matrix | Definition | 1 | 1 |
| Per-interval likelihood factor | Definition | 1 | 1 |

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result | Type | Direct descendants | Total descendants |
|---|---|---|---|
| Martingale Residual | Definition | 1 | 1 |
| Predicted value | Definition | 1 | 1 |
| Risk Set | Definition | 1 | 1 |
| Relative risk ratios | Definition | 1 | 1 |
| Standardized Pearson residual | Definition | 1 | 1 |
| Survival function | Definition | 1 | 1 |
| Censored-data likelihood | Lemma | 1 | 1 |
| Difference in log-odds | Lemma | 1 | 1 |
| Difference of log-hazards between two covariate patterns | Lemma | 1 | 1 |
| General formula for odds ratios in logistic regression | Lemma | 1 | 1 |
| Profile maximum likelihood estimate of each point mass | Lemma | 1 | 1 |
| The whether-a-failure-occurs factor | Lemma | 1 | 1 |
| Exact discretization of the survival factor | Lemma | 1 | 1 |
| Expected mean squares in the one-way random-effects model | Theorem | 1 | 1 |
| Expressions for expit function | Theorem | 1 | 1 |
| Expanded expression for logit | Theorem | 1 | 1 |
| Log-odds as a function of probability | Theorem | 1 | 1 |
| Odds ratios are reversible | Theorem | 1 | 1 |
| Probability as a function of log-odds | Theorem | 1 | 1 |
| Moments of the one-way random-effects model | Theorem | 1 | 1 |
| Mean and variance of residuals | Theorem | 1 | 1 |